Oracle NoSQL Database Cheat Sheet

Oracle NoSQL Database is a distributed, shared-nothing, non-relational database that provides large-scale storage and access to key/value, JSON, and tabular data. It can deliver predictable, low latencies to simple queries at any scale and is designed from the ground up for high availability. For complete documentation, see Oracle NoSQL Database Documentation.

QUICK START TO KVLITE

KVLite is bundled with the Oracle NoSQL Database software.
NOTE: You need a VM instance to run these commands.
Download the .zip file and extract the package. Go to the directory where you unpacked the content.
Secure mode:
```
java -Xmx64m -Xms64m -jar lib/kvstore.jar
kvlite [storagedirsizegb N]
```

Non-secure mode:
```
java -jar lib/kvstore.jar kvlite
[-storagedirsizegb N] -secure-config disable
```

See all the supported parameters in KVLite documentation.

QUICK START TO KVLITE IN A CONTAINER

Pull the image directly from the GitHub Container Registry.
```
docker pull ghcr.io/oracle/nosql:latest-ce

docker tag ghcr.io/oracle/nosql:latest-ce
oracle/nosql:ce
```

Run Oracle NoSQL Database in a Container
```
docker run -d --name=kvlite --hostname=kvlite
--env KV_PROXY_PORT=8080 -p 8080:8080
oracle/nosql:ce
```

Check the Status of the Container
```
docker ps
```

Validate your deployment
```
docker exec -ti kvlite java -jar
lib/kvstore.jar ping -host kvlite -port 5000
```

INSTALL DATA STORE

Download the Oracle NoSQL Database software and extract the package.
```
unzip kv-ee-<latest release>.zip
```

Set the appropriate values for $KVHOME and $KVROOT.
```
export $KVHOME=$HOME/nosql/kv-<latest release>
export $KVROOT=$KVHOME/kvroot
```

Verify the installation
```
java -Xmx64m -Xms64m -jar
$KVHOME/lib/kvclient.jar
```

Verify that the Oracle NoSQL Database processes are running
```
jps -m
```

CONFIGURE DATA STORE

Create the initial bootconfig configuration file on a Storage Node.
Using password file:
```
java -jar $KVHOME/lib/kvstore.jar
makebootconfig -root kvroot -host $KVHOST -
harange 5005,5050 -port 5000 -store-security
configure -pwdmgr pwdfile -capacity 1 -
storagedir <storage path> -storagedirsize
<n>_GB
```
Using wallet:
```
java -jar $KVHOME/lib/kvstore.jar
makebootconfig -root kvroot -host $KVHOST -
harange 5005,5050 -port 5000 -store-security
configure -pwdmgr wallet -capacity 1 -
storagedir <storage path> -storagedirsize
<n>_GB
```

NOTE: For non-secure data store, include `store-security` parameter as `none`.

Start the Oracle NoSQL Database Storage Node Agent (SNA)
```
nohup java -Xmx64m -Xms64m \
-jar $KVHOME/lib/kvstore.jar start -root
$KVROOT &
```

Check if the SNA is running on the Storage Node
```
ssh <Storage node>
java -Xmx64m -Xms64m -jar
$KVHOME/lib/kvstore.jar ping -host $KVHOST -
port 5000 -security
$KVROOT/security/client.security
```

For non-secure data store, remove `-security` parameter

Stop a Storage Node
```
java -Xmx64m -Xms64m -jar
$KVHOME/lib/kvstore.jar stop -root $KVROOT
```

Clear KVROOT directory
```
rm -rf $KVROOT
```
See Configuring a single region data store.

ADMIN CLI COMMANDS

Start the Admin Command Line Interface (CLI) utility
```
java -jar $KVHOME/lib/kvstore.jar runadmin -
host $KVHOST -port 5000 -security
$KVROOT/security/ client.security
```

Name your data store
```
configure -name <store name>
```

Create a Zone
```
plan deploy-zone -name <zone name>
-rf <replication factor> [-type [primary |
secondary]][-arbiters | -no-arbiters ]
[-json ][-master-affinity | -no-master-
affinity][-plan-name <name>] [-wait] [-
noexecute] [-force]
```

View Store topology
```
show topology
```

Deploy a Storage Node (<sn>)
```
plan deploy-sn -zn <zone name> -host <sn> -
port 5000 -wait
```

Create admin process on the Storage Node
```
plan deploy-admin -sn <sn> -wait
```

Create a Storage Node Pool
```
pool create -name <pool name>
pool join -name <pool name> -sn <sn>
```

Calculate number of partitions
```
(Total number of disks hosted by the Storage
Nodes / Replication Factor) * 10
```

Create Topology name
```
topology create -name <topology name> -pool
<pool name> -partitions <partition count>
```

Create and Deploy Replication Nodes
```
plan deploy-topology -name <topology name> -
wait
```

Check plan status
```
show plans
```

Verify data store
```
verify configuration
ping <sn>
```

Create an admin user
```
execute 'CREATE USER root IDENTIFIED BY
\"password\" ADMIN'
```

Create password file to store credentials for admin user (root)
Run from an Oracle NoSQL Database instance.
```
java -Xmx64m -Xms64m -jar
$KVHOME/lib/kvstore.jar securityconfig pwdfile
create -file $KVROOT/security/login.passwd
java -Xmx64m -Xms64m -jar
$KVHOME/lib/kvstore.jar securityconfig pwdfile
secret -file $KVROOT/security/login.passwd -
set -alias root
<Supply the password for admin user and re-
enter the password to confirm>
```

Create the wallet to store credentials for the admin user (root)
```
java -Xmx64m -Xms64m -jar
KVHOME/lib/kvstore.jar securityconfig wallet
create -dir $KVROOT/security/login.wallet
java -Xmx64m -Xms64m -jar
KVHOME/lib/kvstore.jar securityconfig wallet
secret -dir $KVROOT/security/login.wallet -set
-alias root
<Supply the secret value for admin user and
re-enter the value to confirm>
```

START SQL SHELL

Run from an Oracle NoSQL Database instance.
```
java -jar $KVHOME/lib/sql.jar -helper-hosts
node01:5000 -store kvstore
```

See all the supported parameters in Running the SQL Shell.

USER MANAGEMENT

Run the commands from SQL shell.

Create user
```
CREATE USER user_name (IDENTIFIED EXTERNALLY |
IDENTIFIED BY password [PASSWORD EXPIRE |
PASSWORD LIFETIME duration_time_unit])
[ACCOUNT LOCK|UNLOCK] [ADMIN]
```

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Manage user-defined roles

```
CREATE ROLE role_name
DROP ROLE role_name
SHOW ROLES
```

Grant privileges to a user-defined role

```
GRANT { grant_roles | grant_system_privileges|
grant_object_privileges }
grant_roles ::= role [, role]... TO { USER
user | ROLE role }
grant_system_privileges ::= {system_privilege
| ALL PRIVILEGES}[,{system_privilege | ALL
PRIVILEGES}]...TO role
grant_object_privileges ::={object_privileges|
ALL [PRIVILEGES]}[,{object_privileges| ALL
[PRIVILEGES]}]...ON object TO role

Examples:
CREATE ROLE manager
GRANT WRITE_ANY TO manager
CREATE ROLE employee
GRANT READ_ANY TO employee
GRANT employee TO ROLE manager
GRANT manager TO USER Kate
GRANT READ_TABLE on T1 TO Kate
```

ORACLE NOSQL DATABASE PROXY

Run from an Oracle NoSQL Database instance.

Secure mode:

```
java -jar $KVHOME/lib/httpproxy.jar -storeName
<store name> -helperHosts $KVHOST [-hostname
<proxy_host>] [-httpsPort <proxy_https_port>]
-storeSecurityFile
$KVROOT/security/proxy.login -sslCertificate
certificate.pem -sslPrivateKey key-pkcs8.pem -
sslPrivateKeyPass <privatekey_password> [-
verbose true]
```

Non-secure mode:

```
java -jar $KVHOME/lib/httpproxy.jar -storeName
<store name> -helperHosts $KVHOST [-hostname
<proxy_host>] [-httpPort <proxy_http_port>]
```

See all the supported parameters in Configuring the Proxy.

ORACLE NOSQL DATABASE MIGRATOR

Download the Oracle NoSQL Database Migrator utility from the Oracle NoSQL Downloads page.

Identify the Source and Sink. See Supported Sources and Sinks.
Prepare the configuration file (JSON format) with the identified Source and Sink details. See Source Configuration Templates and Sink Configuration Templates.

Run the Migrator utility from the command prompt to perform migration by passing the configuration file

```
./runMigrator --config <complete path to the
JSON config file>
```

Run the Migrator utility from the command prompt to generate the configuration file interactively and then perform migration

```
./runMigrator
```

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